

The growth-record studio

Teacher guide and worked answers

Set up a consistent growth record and use dated repeated measurements to describe change over time.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

[_passsb/inputs/Build SOW 2026-2027.xlsx](#) | BUILD Weekly - Summer | C34

Set up and observe a plant-growth investigation over time (ICT to record).

Develops the C34 setup and recording skills, then analyses supplied dated evidence within 40 minutes. An optional real class investigation continues over days; this single lesson does not claim to complete ten days of live observation.

Prior learning

A fair plan compares similar plants with one intended change.

Height and leaf condition provide complementary evidence.

Success evidence

Use a plant ID, date, unit and fixed height rule.

Read change across the supplied dates.

Retain variation and distinguish missing from zero.

Equipment

Printed shared evidence and one chosen pupil route; pencil; optional ruler; projector or offline HTML.

Optional established bean seedlings, labelled pots and trays; centimetre ruler; optional spreadsheet.

Preparation

Print the shared evidence, one response route and exit page. Routes are alternatives, not a workload ladder.

Read the worked answers. Prepare an oral, pointing or adult-scribed route where useful.

The dated May 2027 table is constructed teaching evidence, not historical observations. All six plants start at 3 cm. Group A receives 5 mL daily and B 15 mL in the fictional plan; do not treat these as universal care instructions.

Optional setup: label a real plant, take one baseline reading and arrange future observation dates with an adult. Keep that record separate from the supplied data. The full ten-day analysis uses the provided table today.

Practical care

Use adult-selected non-toxic plants for any optional baseline; no tasting. Wash hands after handling plants or compost.

Keep small water volumes in trays away from devices and wipe spills. An adult checks ongoing plant care daily and records any necessary adjustment.

Ready to teach alternative

All core reasoning uses the supplied evidence and can be completed in 40 minutes without a live practical.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	A record should survive a change of observer
2	Retrieval	2-5	Retrieve the measurement rule
3	I do	5-12	I set up the record before comparing
4	I do	Within stage	I read time before height
5	I do	Within stage	I preserve a missing observation
6	We do	12-16	The timeline evidence desk
7	We do	16-20	Keep the differences visible
8	Hinge	20-23	B1 starts at 3 cm and ends at 11 cm.
9	You do	23-25	Choose your science route
10	You do	25-35	Make your own evidence record
11	You do	Within stage	Check what the evidence can show
12	Review	35-38	Lundy loop: improve the shared account
13	Review	Within stage	Keep the useful change
14	Exit	38-40	Three final checks

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening A record should survive a change of observer

State explicitly that animation steps through constructed records; no real plant grows ten days during this lesson.

2 Retrieval Retrieve the measurement rule

1 soil surface. 2 cm. 3 track the same plant over time and preserve repeats.

3 I do I set up the record before comparing

Think aloud: "I record B1, date, cm and 3 cm starting height. Later 11 cm does not mean 11 cm growth: 11 minus 3 is 8 cm." Demonstrate one real baseline if ready, or use the labelled worked model.

4 I do I read time before height

Shares I do time. Think aloud: "Day 3 to day 7 is four days, not three. The graph gives that interval more width. At day 10 the means are 6 and 11 cm." Mean is an optional summary, explained as total divided by three; core pupils read individual rows.

5 I do I preserve a missing observation

Shares I do time. Think aloud: “No one measured plant X, so I cannot write zero. I mark missing. Our supplied six-plant table happens to be complete.”

6 We do The timeline evidence desk

W1: B1 goes from 3 to 11 cm; increase 8 cm. Have a pupil choose the date before the other reads the value. Step the model only after predicting.

7 We do Keep the differences visible

W2: retain all three; natural variation is useful. At day 10 B plants are green with 5-6 leaves; A plants have 2-3 leaves and A3 slightly droops. Heights alone do not prove overall health or a universal best water amount.

Reveal: Describe the pattern and its limits; keep every reading.

8 Hinge B1 starts at 3 cm and ends at 11 cm.

Correct second option. Use a difference bar or counted steps if needed.

A. 11 cm - That is the final height, not the change.

B. 8 cm - 11 minus 3 equals 8 cm.

C. 0 cm because one other plant was not measured - B1 has both measurements; a separate missing value does not erase them.

9 You do Choose your science route

Set ONE route. All routes analyse the supplied dates within this lesson. Optional setup means one baseline today plus an explicitly future schedule. Standard graph values B1: (0,3), (3,5), (7,8), (10,11). Do not require mean calculation for core BUILD.

10 You do Make your own evidence record

All routes analyse the supplied dates within this lesson. Optional setup means one baseline today plus an explicitly future schedule. Standard graph values B1: (0,3), (3,5), (7,8), (10,11). Do not require mean calculation for core BUILD.

11 You do Check what the evidence can show

Shares independent time. Ask for a reason, not just a correct word. All routes analyse the supplied dates within this lesson. Optional setup means one baseline today plus an explicitly future schedule. Standard graph values B1: (0,3), (3,5), (7,8), (10,11). Do not require mean calculation for core BUILD.

12 Review Lundy loop: improve the shared account

Listen to a selected pupil revision and visibly adopt a scientifically accurate contribution. Offer private, written or partner feedback.

Reveal: Use change from baseline, keep repeats and leaf evidence, and show missing records as missing.

13 Review Keep the useful change

Shares review time. Name how a pupil contribution changed a label, method or conclusion. Do not rank pupils.

Reveal: Use change from baseline, keep repeats and leaf evidence, and show missing records as missing.

14 Exit Three final checks

Collect brief individual evidence. Use the answer key and re-teach the specified misconception if needed.

Answers and assessment

Retrieval 1-3

1 soil surface. 2 cm. 3 compare the same plant across dates and retain individual variation.

We do W1-W2

W1 3 cm to 11 cm, increase 8 cm. W2 distinct individuals vary; keep 11,12,10 rather than manufacture equality. Add leaf notes and limit conclusion to these conditions.

Hinge

Second option correct: 8 cm. 11 is final height; missing data in another record is not zero for B1.

S1-S5

S1 3 cm. S2 11 cm. S3 8 cm. S4 e.g. B plants green with 5-6 leaves, A3 slightly drooping. S5 missing with a reason.

M1-M4

M1 3 cm, 11 cm, increase 8 cm. M2 points (day,height) (0,3),(3,5),(7,8),(10,11), joined in order with units; time spacing is unequal. M3 in these constructed conditions B ended taller (10-12 cm versus A 5-7 cm) with more leaves, while repeats varied. M4 one species, pot size, stage, short interval and limited sample cannot cover all plants/conditions.

T1-T4

T1 B final heights 11,12,10 cm from 3 cm baselines, so increases are 8,9,7 cm. T2 A 5-7 cm; B 10-12 cm; ranges do not overlap here. T3 any two: small sample, one species/stage/pot size, limited interval, specific growing conditions, height not full health. T4 mark missing and give date/reason; retain all actual observations; take a later reading with its real date rather than invent the missed one.

Exit E1-E3

E1 starting measurement used for comparison. E2 8 cm. E3 no.

R1-R2

Responses vary. Credit a scientific revision, evidence-based defence or relevant next question. A private or oral response has equal value.

Teaching and assessment notes

40 minutes: opening 2, retrieval 3, I do 7, We do 8, hinge 3, independent 12, review 3, exit 2. Zero-minute slides share the named stage time.

BUILD uses the supplied Year 3 science anchor with age-respectful contexts for older pupils. Supported, standard and stretch are selectable routes within the pathway.

Interactive We do: predict first, test against the controlled SVG model or supplied evidence, then explain or revise. Shared W tasks offer the equivalent paper discussion. Animation is a teaching representation, not filmed experimental evidence.

Keep constructed evidence separate from actual class observations. Do not invent measurements or require internet research.

The exact group means are provided for the interactive summary; no rounding is required. Mean A: 3,4,5,6; mean B: 3,5,8,11 cm. Keep individual repeats available alongside the summary.

Secure core: consistent record identifiers and units, correct change from baseline, and an honest statement about missing or variable readings.

Access and responsive teaching

Supported

Short choices, word bank and pointing or adult-scribed reasons; same core scientific goal.

Standard

Make a short evidence-based explanation or record using the supplied information.

Stretch

Apply the core idea to a changed case, evaluate evidence or identify a limit.

Regulation

Preview the sequence. Offer seated observation, quiet paired rehearsal, private feedback and a pass from public sharing. Routes respond to current access needs, not a diagnosis.

Misconceptions to check

The final height is the amount of growth.

Growth in height is final minus starting height.

Check: Compare 11 cm final with 3 cm baseline.

No reading means zero growth or zero height.

A missing reading cannot be replaced by zero.

Check: Ask whether a measurement was actually made.

Each repeat must have the same value.

Individual plants vary; keep the observed differences.

Check: Compare B1-B3 on day 10.

Word	Meaning
baseline	The starting measurement for comparison.
interval	The time between observations.
variation	Differences between individual readings.
missing	A value that was not recorded; not the same as zero.

Scientific references

[Department for Education: science programmes of study](#)

Checked 8 September 2026. Year 3 plants, rocks, light and forces underpin the SOW. Accessible teaching models are adapted to the named lesson objective.

[Science and Plants for Schools: plant biology animations](#)

Checked 8 September 2026. Scientific background for our original simplified diagrams: photosynthesis, growth and transport. No external animation is required or embedded.

[Royal Horticultural Society: sowing seeds indoors](#)

Checked 8 September 2026. Germination uses stored reserves and requires appropriate water, oxygen and temperature; seedling care differs from starting germination.